Test for the mean. Population variance known
Data scientist salary

Background You are given the data from the lesson
Task 1 Test the null hypothesis at 10% significance

Solution:

Dataset
$ 117,313
$ 104,002
$ 113,038
$ 101,936
$ 84,560
$ 113,136
$ 80,740
$ 100,536
$ 105,052
$ 87,201
$ 91,986
$ 94,868
$ 90,745
$ 102,848
$ 85,927
$ 112,276
$ 108,637
$ 96,818
$ 92,307
$ 114,564
$ 109,714
$ 108,833
$ 115,295
$ 89,279
$ 81,720
$ 89,344
$ 114,426
$ 90,410
$ 95,118
$ 113,382

Sample mean	$ 100,200
Population std	$ 15,000
Standard error	$ 2,739
Sample size	30
Z-score	-4.67

H_0 (Glassdoor data) $ 113,000
Two-sided test

- z = the z-score
- x = the specific data value you are looking at
- μ (or $\bar{x}$) = the mean (the average value of all data points)
- σ (or s) = the standard deviation (a measure of how spread out the numbers are)

- A **positive z-score** means the value is **above** the average.
- A **negative z-score** means the value is **below** the average.
- A **z-score of 0** means the value is exactly **equal** to the average.

Task 1 *Two-sided test*

$z_{0.05}$ 1.65

4.67 > 1.65, therefore, we reject the null hypothesis
There was no need for the test as we have already rejected it lower significance levels. You could simply say that you would reject the null hypothesis.

For a **population**:

$$z = \frac{x - \mu}{\sigma}$$

For a **sample**:

$$z = \frac{x - \bar{x}}{s}$$